

Harthik Royal Mallichetty

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PROFESSIONAL SUMMARY

Analytics Engineer and May 2026 MSBA graduate with production experience building data transformation pipelines, metric layers, and self-service reporting infrastructure. Built Signal, a financial intelligence platform where dbt 3-layer architecture computes 65 metrics across 70 S&P 500 companies with Airflow orchestration on GCP. Previously built a production ML and reporting system at Prolifics used by 200 developers daily, still running in 2025. Proficient in Python, SQL, dbt, Apache Airflow, PostgreSQL, Tableau, Power BI, and GCP.

EDUCATION

The University of Texas at Dallas, Richardson, Texas

Master of Science in Business Analytics and Artificial Intelligence – Scholar With Distinction

May 2026

GPA 3.81

GITAM University, Hyderabad, India

April 2024

Bachelor of Technology in Computer Science and Business Systems

GPA 3.73

PROFESSIONAL EXPERIENCE

Prolifics – Data & Analytics Intern | Hyderabad, India

Feb 2023 – Jan 2024

- Identified data quality gaps across 100K+ PostgreSQL records, built complex SQL pipelines to clean and transform raw sprint data into model-ready tables, and deployed a TensorFlow prediction model via Flask API into the company portal; system used by 200 developers daily and still running in production 2025.
- Automated ETL workflows processing 50K+ rows/day using Python, reducing reporting cycle time by 40% and improving downstream accuracy by 25%; built and maintained Power BI dashboards tracking operational KPIs for Finance, DevOps, and QA stakeholders.

HeiDigi – Digital Marketing Analyst Intern | Hyderabad, India

Mar 2022 – Aug 2022

- Built automated Tableau dashboards pulling Google Ads performance data, surfacing KPIs that improved ROI by 20% and CTR by 18%; translated complex campaign data into clear reporting outputs for non-technical leadership.
- Designed and executed A/B tests applying hypothesis testing and statistical inference; communicated findings to CEO-level stakeholders driving 15% revenue impact increase.

PROJECTS

Signal – Financial Intelligence | Python, FastAPI, pgvector, dbt, Airflow, Voyage AI, GPT-4o-mini, GCP

May 2026

- Designed dbt 3-layer transformation architecture (Staging, Intermediate, Marts) computing 65 financial metrics per company across 70 S&P 500 companies; marts layer serves as the sole API data source, ensuring no raw table queries hit production endpoints.
- Built 8-module Python ingestion pipeline pulling SEC EDGAR filings and Polygon.io market data with dual-key failover, rate limiting, and failed ingestion logging so the pipeline never crashes on a single company failure; orchestrated via Airflow DAGs on GCP Cloud SQL.

Quant Risk Desk | Python, SQL, dbt, Apache Airflow, PostgreSQL, Docker, Tableau

April 2026

- Built 9-task Airflow DAG ingesting 21K+ daily records across 11 S&P 500 sector ETFs; implemented 18 dbt data quality tests enforcing mathematical integrity constraints with zero violations across all downstream reporting layers.
- Designed 3-layer dbt transformation computing VaR/CVaR (95%/99%), Sharpe/Sortino, rolling Beta, and Macro Regime Engine using Fed Funds Rate, CPI, VIX, and yield curve signals; delivered Tableau dashboard with sector KPI scorecards, VaR envelopes, drawdown charts, and correlation heatmaps.

Global Supply Chain Intelligence | Python, SQL, XGBoost, LightGBM, SHAP, NetworkX, Plotly

Mar 2026

- Designed scalable data pipeline integrating 3 heterogeneous sources across 78K weekly records and 500 SKUs; engineered feature set with SHAP TreeExplainer attributing 59% of model weight to disruption signals.
- Built CUSUM and Mahalanobis anomaly detection achieving 100% disruption recall with 8-week lead time; executed Monte Carlo simulation with 10K runs per SKU across 3 stress scenarios and delivered 6-tab interactive Plotly dashboard for non-technical stakeholder review.

SKILLS

Programming and Databases: Python (Pandas, NumPy, scikit-learn, XGBoost, SHAP, SciPy, Plotly), SQL (CTEs, Window Functions, Self-Joins, Rolling Aggregations, Query Optimization), R, PostgreSQL, DuckDB, MySQL, Git/GitHub

Data Transformation and Engineering: dbt (Staging/Intermediate/Marts, custom tests, data quality enforcement), Apache Airflow (DAGs), ETL Pipeline Automation, Dimensional Modeling, Star Schema Design, Data Validation and Quality Testing

BI and Visualization: Tableau, Power BI (DAX, Data Modeling, KPI Engineering), Streamlit, Plotly, Excel (PivotTables, Advanced Functions, Forecasting), Self-Service Dashboard Design, Data Storytelling

Cloud and Tools: GCP (Cloud Run, Cloud SQL), Docker, GitHub Actions, FastAPI, n8n, SEC EDGAR API, Polygon.io

Statistical Methods: A/B Testing, Hypothesis Testing, Anomaly Detection (CUSUM, Mahalanobis), Time-Series Forecasting (ARIMA, Prophet), Regression Analysis, Monte Carlo Simulation

ACTIVITIES

Cloud Analytics Lead – GDSC, GITAM Hyderabad

Apr 2022 – Apr 2024

- Built GCP and BigQuery pipelines analyzing behavioral data to quantify investment impact on college admission rates.